Example 3.52 Solve $2\sin^2 x + \sin^2 2x = 2$

Solution:

$2\sin^2 x + \sin^2 2x = 2 \Rightarrow 2\sin^2 x + (2\sin x \cos x)^2 = 2$

$\cos^2 x \left(2\sin^2 x - 1\right) = 0$

Either $\cos x = 0$ or $\sin^2 x = \dfrac{1}{2}$

$\cos x = 0$

$\Rightarrow x = (2n+1)\dfrac{\pi}{2}, n \in \mathbb{Z}$

$\sin^2 x = \dfrac{1}{2} = \sin^2 \dfrac{\pi}{4}$

$x = n\pi \pm \dfrac{\pi}{4}, n \in \mathbb{Z}$ How

Thus the solution is $x = (2n+1)\dfrac{\pi}{2}, \quad x = n\pi \pm \dfrac{\pi}{4}, n \in \mathbb{Z}.$

Example 3.53 Prove that for any a and b , $-\sqrt{a^2+b^2} \leq a\sin\theta + b\cos\theta \leq \sqrt{a^2+b^2}$

Solution:

Now, $a\sin\theta + b\cos\theta = \sqrt{a^2+b^2}\left[\dfrac{a}{\sqrt{a^2+b^2}}\sin\theta + \dfrac{b}{\sqrt{a^2+b^2}}\cos\theta\right]$

$= \sqrt{a^2+b^2}\left[\cos\alpha\sin\theta + \sin\alpha\cos\theta\right]$

where $\cos\alpha = \dfrac{a}{\sqrt{a^2+b^2}}, \ \sin\alpha = \dfrac{b}{\sqrt{a^2+b^2}}$

$= \sqrt{a^2+b^2} \ \sin(\alpha+\theta)$

Thus, $|a\sin\theta + b\cos\theta| \leq \sqrt{a^2+b^2}$

Hence, $-\sqrt{a^2+b^2} \leq a\sin\theta + b\cos\theta \leq \sqrt{a^2+b^2}.$

Example 3.54 Solve $\sqrt{3}\sin\theta - \cos\theta = \sqrt{2}$

Solution:

$\sqrt{3}\sin\theta - \cos\theta = \sqrt{2}$

Here $a = -1; b = \sqrt{3}; c = \sqrt{2}; r = \sqrt{a^2+b^2} = 2.$

Thus, the given equation can be rewritten as

$\dfrac{\sqrt{3}}{2}\sin\theta - \dfrac{1}{2}\cos\theta = \dfrac{1}{\sqrt{2}}$

$\sin\theta\cos\dfrac{\pi}{6} - \cos\theta\sin\dfrac{\pi}{6} = \sin\dfrac{\pi}{4}$

$\sin\left(\theta - \dfrac{\pi}{6}\right) = \sin\dfrac{\pi}{4}$

$\theta - \dfrac{\pi}{6} = n\pi \pm (-1)^n \dfrac{\pi}{4}, n \in \mathbb{Z}$

Thus, $\theta = n\pi + \dfrac{\pi}{6} \pm (-1)^n \dfrac{\pi}{4}, n \in \mathbb{Z}.$